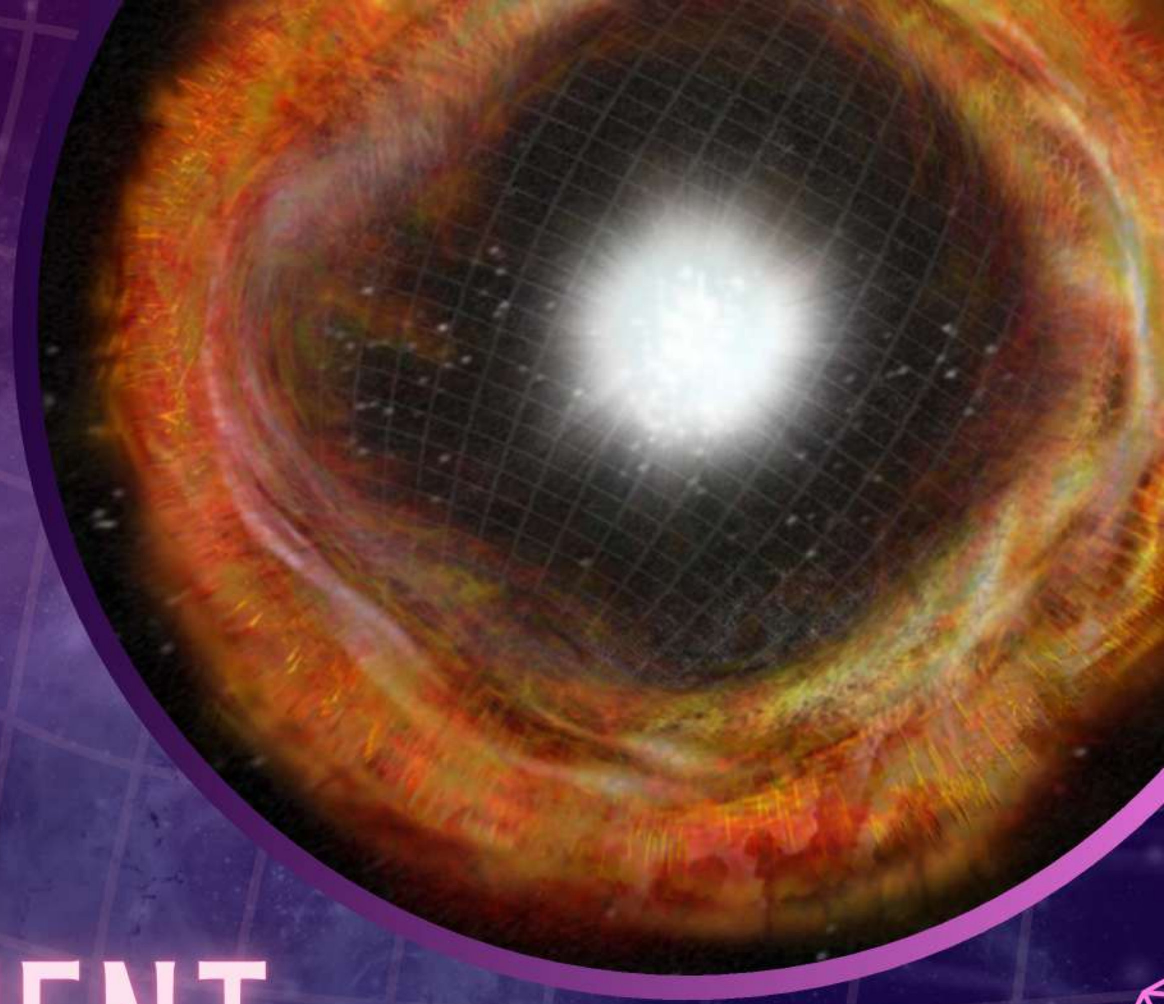
OzGrav + KAGRA ECR School

1 July 2026



Introduction to Poorly-Modelled Transient Gravitational Waves (Bursts)

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OzGrav ARC CENTRE OF EXCELLENCE FOR GRAVITATIONAL WAVE DISCOVERY

LIGO VIRGO KAGRA

Talk Overview

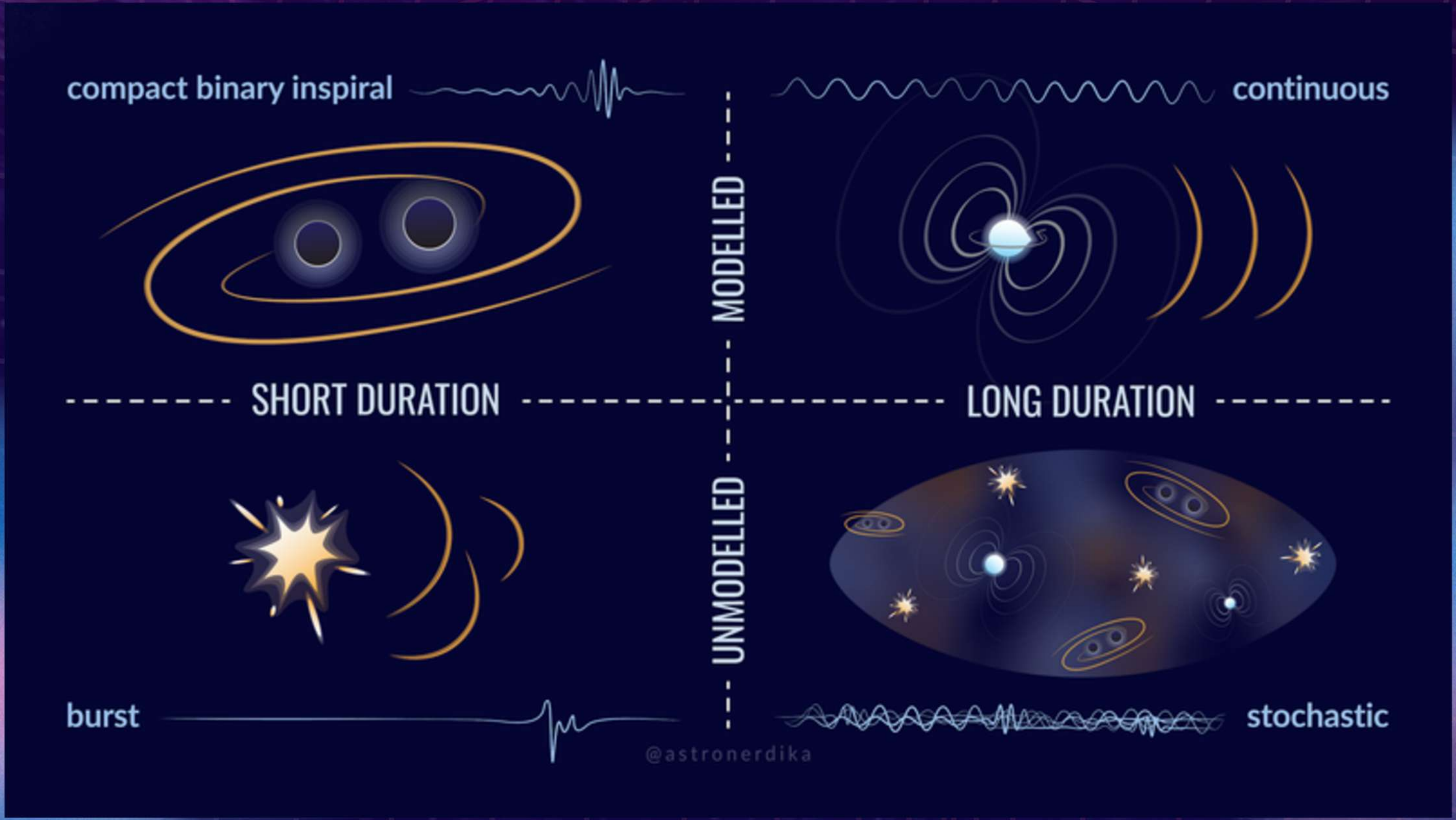
INTRODUCTION

- Types of transient GW sources (bursts)
- Excess power & Instrumental glitches

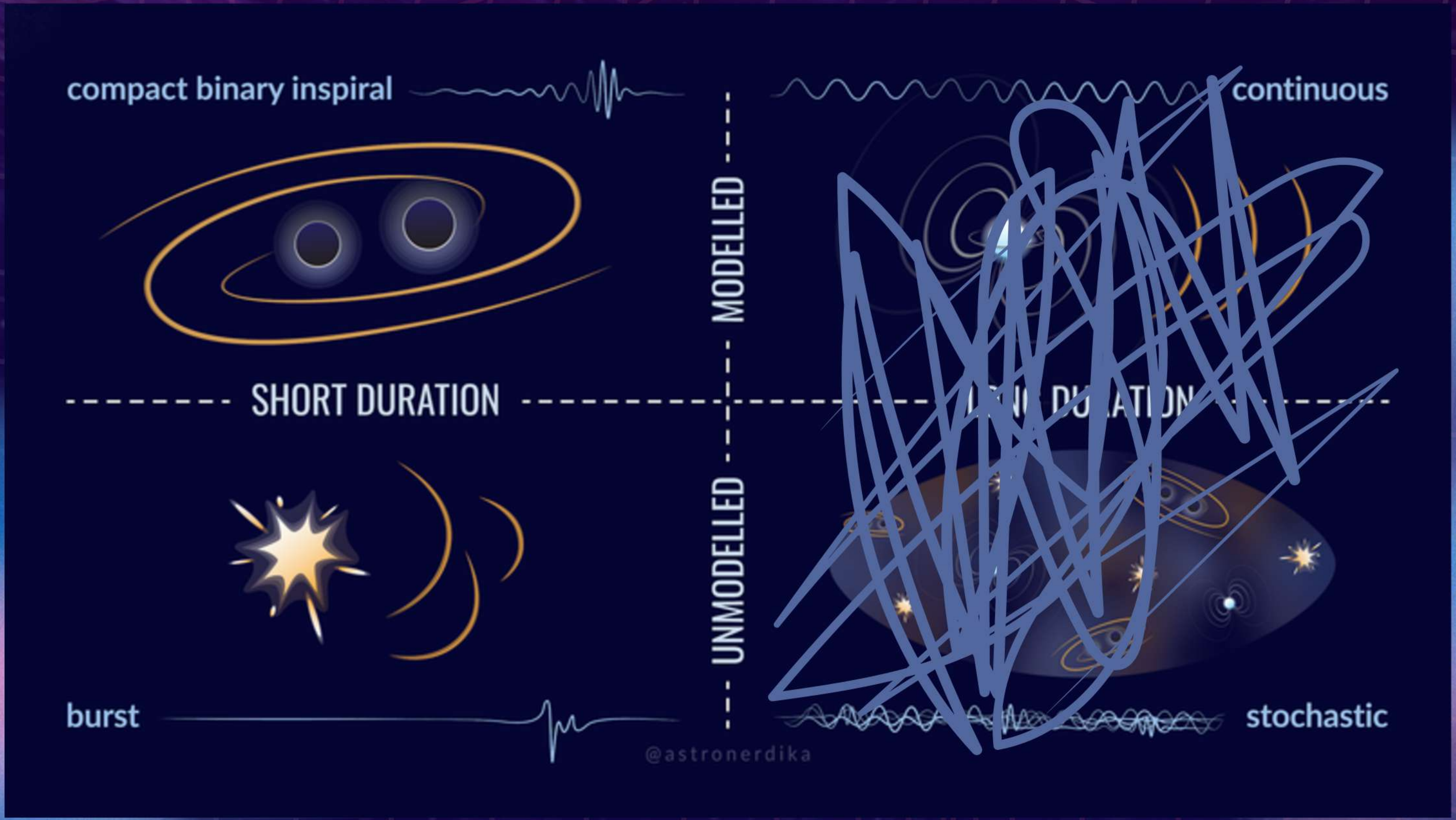
GW BURST ANALYSIS

- **Modelled** vs. **unmodelled** searches
- **Detection statistics** and **false alarms**
- LVK burst search pipelines & their **roles in GW astronomy**

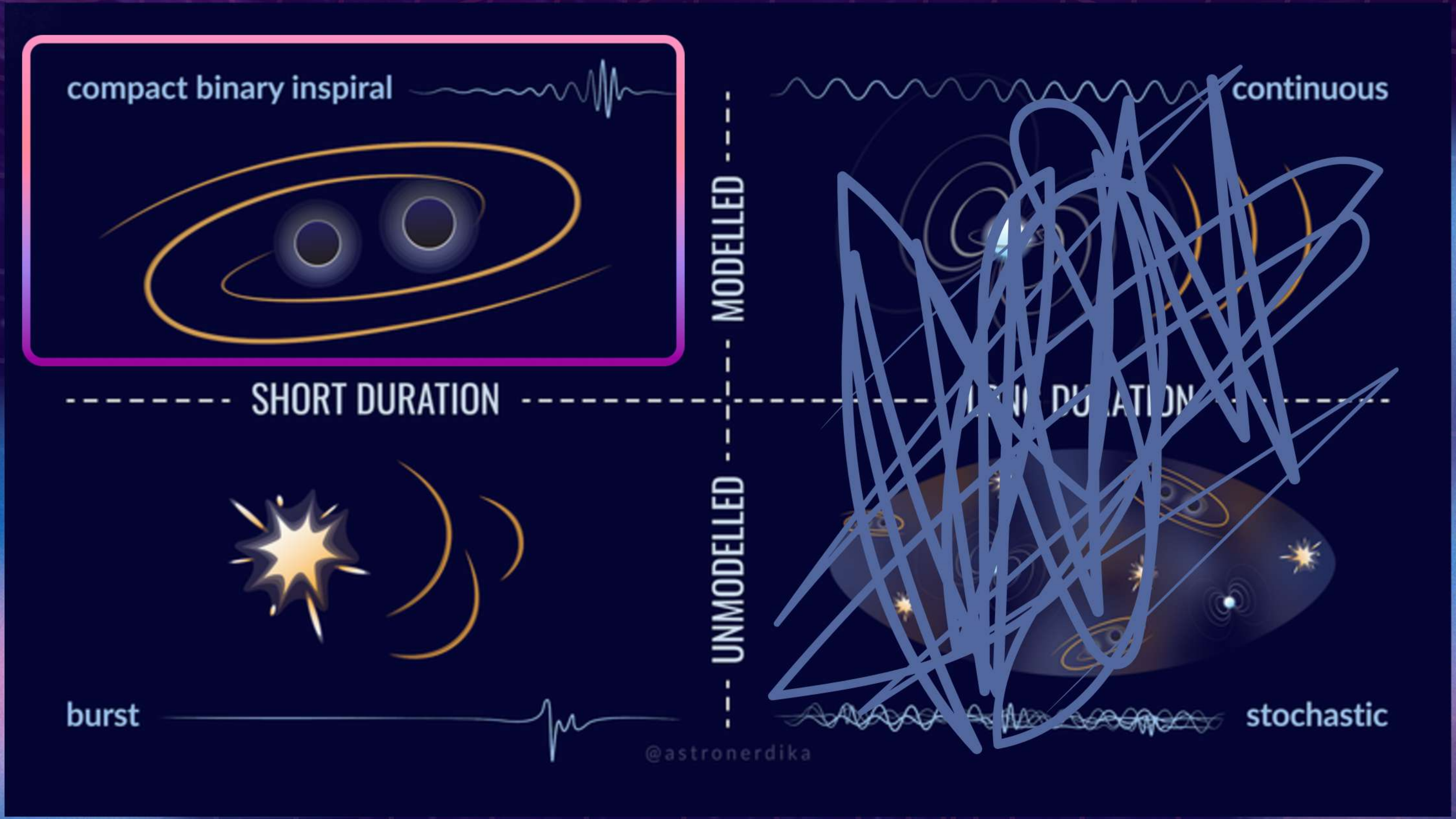
Types of
Gravitational Waves



Types of
Gravitational Waves

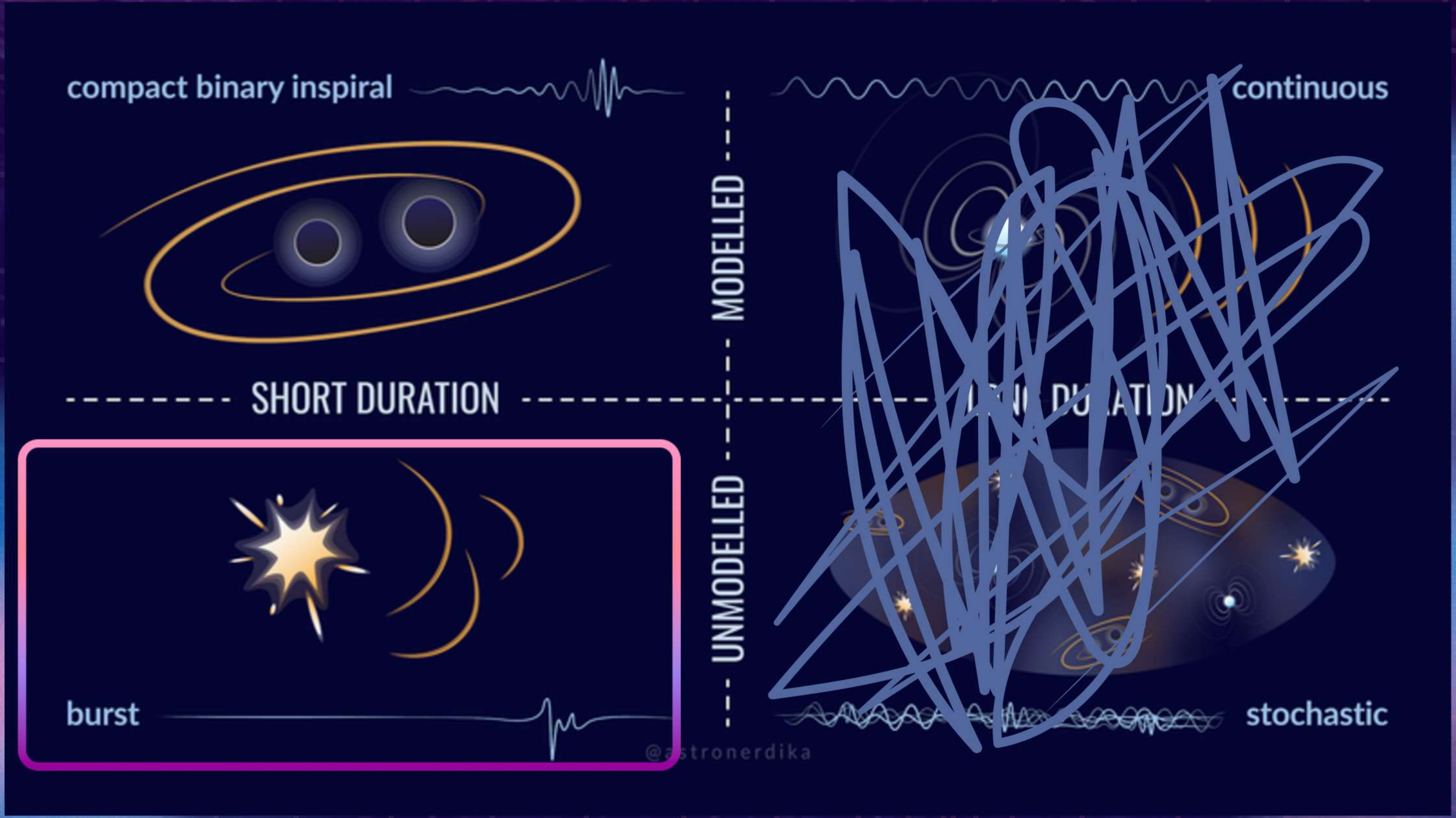


Types of **Gravitational Waves**



$\mathcal{O}(10^2)$
events
detected

Types of
Gravitational Waves

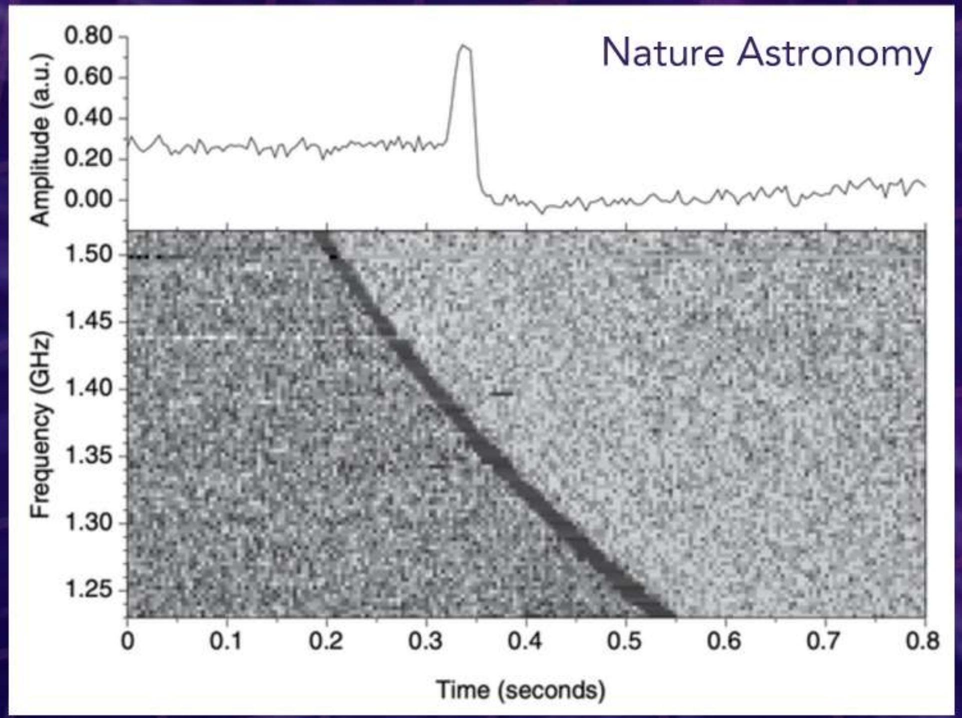


Focus of
this talk

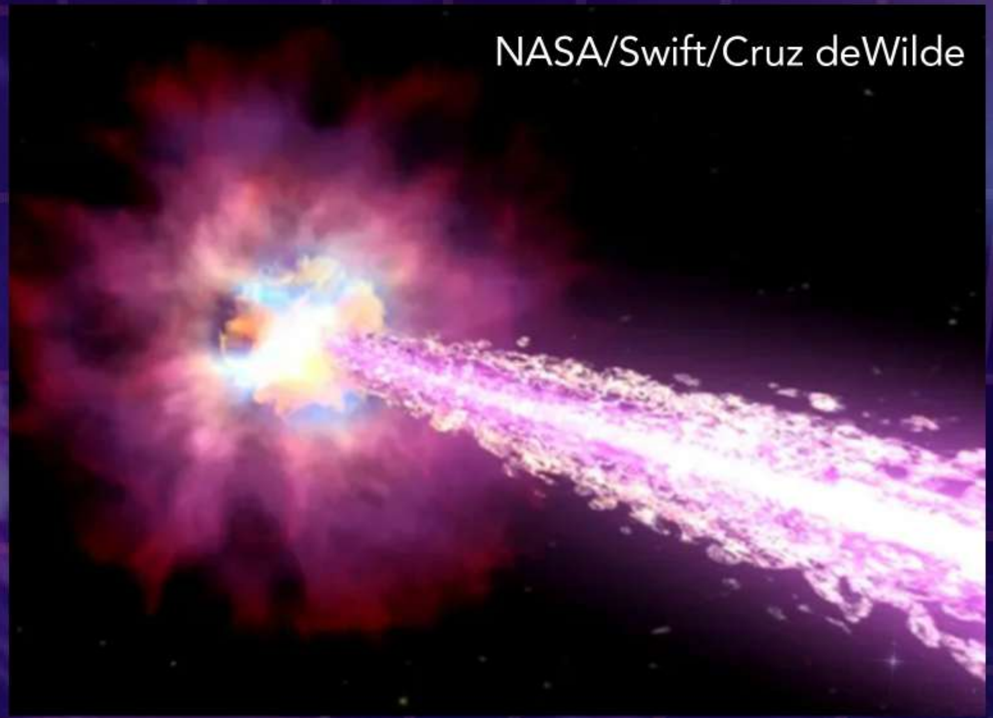
Why search for **Unmodelled bursts**?

Multi-messenger astronomy for poorly-modelled sources

Core-collapse supernovae (CCSNe)



Fast radio bursts



Gamma ray bursts

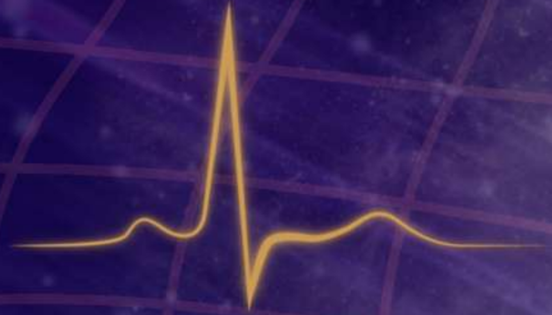
The unknown unknown:
Potential disovery of new, obscure sources

What is in the **GW detector data**

$$\underbrace{\vec{s}}_{\text{Data}} = \underbrace{\vec{n}}_{\text{Noise}} + \underbrace{\vec{h}(\vec{\theta})}_{\text{Signal}}$$

$$\vec{n} = \vec{n}_G + \vec{g}$$

$\vec{n}_G$: Stationary, Gaussian noise



$\vec{g}$: Non-stationary, non-Gaussian, **Instrumental glitch(es)**

What is in the GW detector data

$$\vec{s} = \vec{n} + \vec{h}(\vec{\theta})$$

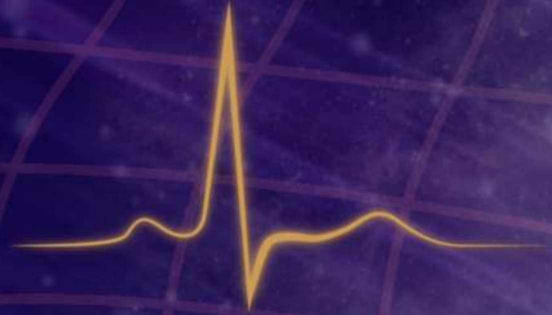
Data **Noise** **Signal**

EXCESS POWER

$$\vec{n} = \underbrace{\vec{n}_G + \vec{g}}$$

$\vec{n}_G$

Stationary,
Gaussian noise



$\vec{g}$

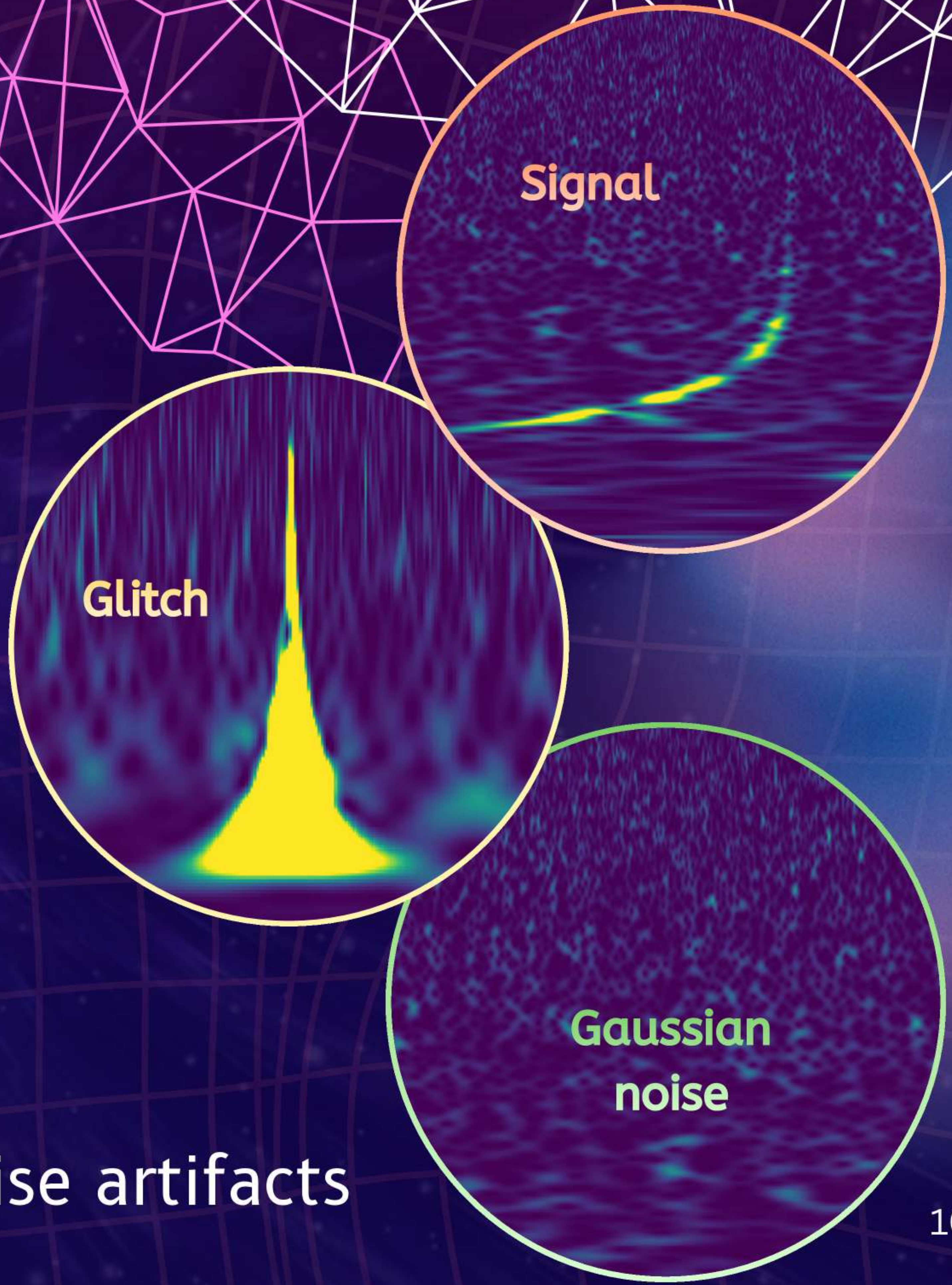
Non-stationary, non-Gaussian,
Instrumental glitch(es)

Localised excess power

a.k.a bursts

- Stands out from the stationary and Gaussian noise floor
- Localised in time [$\lesssim$ 1 second**]
- Span a broad frequency range

But... there exists an impostor



INSTRUMENTAL GLITCHES

Non-gaussian, non-stationary, transient noise artifacts

Instrumental glitches

can mimic / mask burst signals

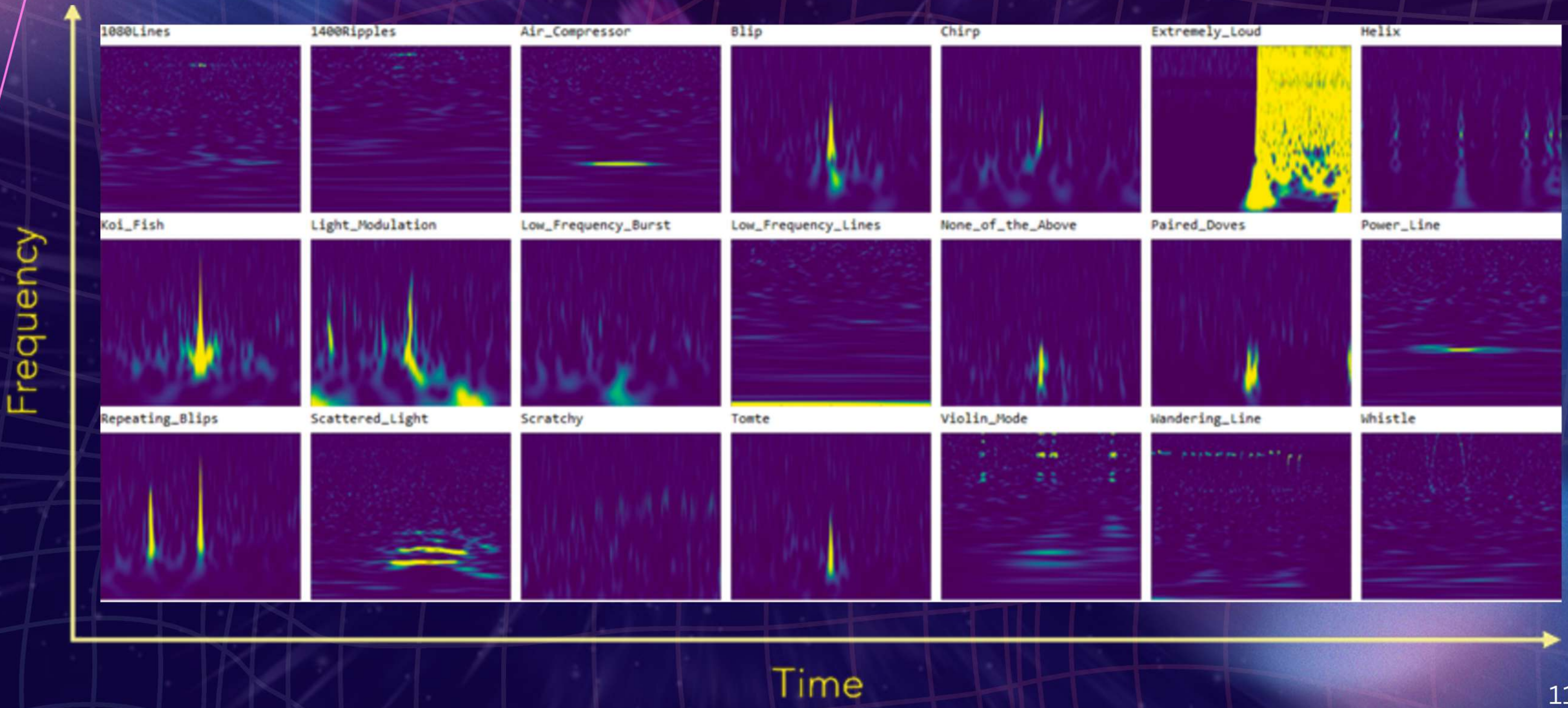
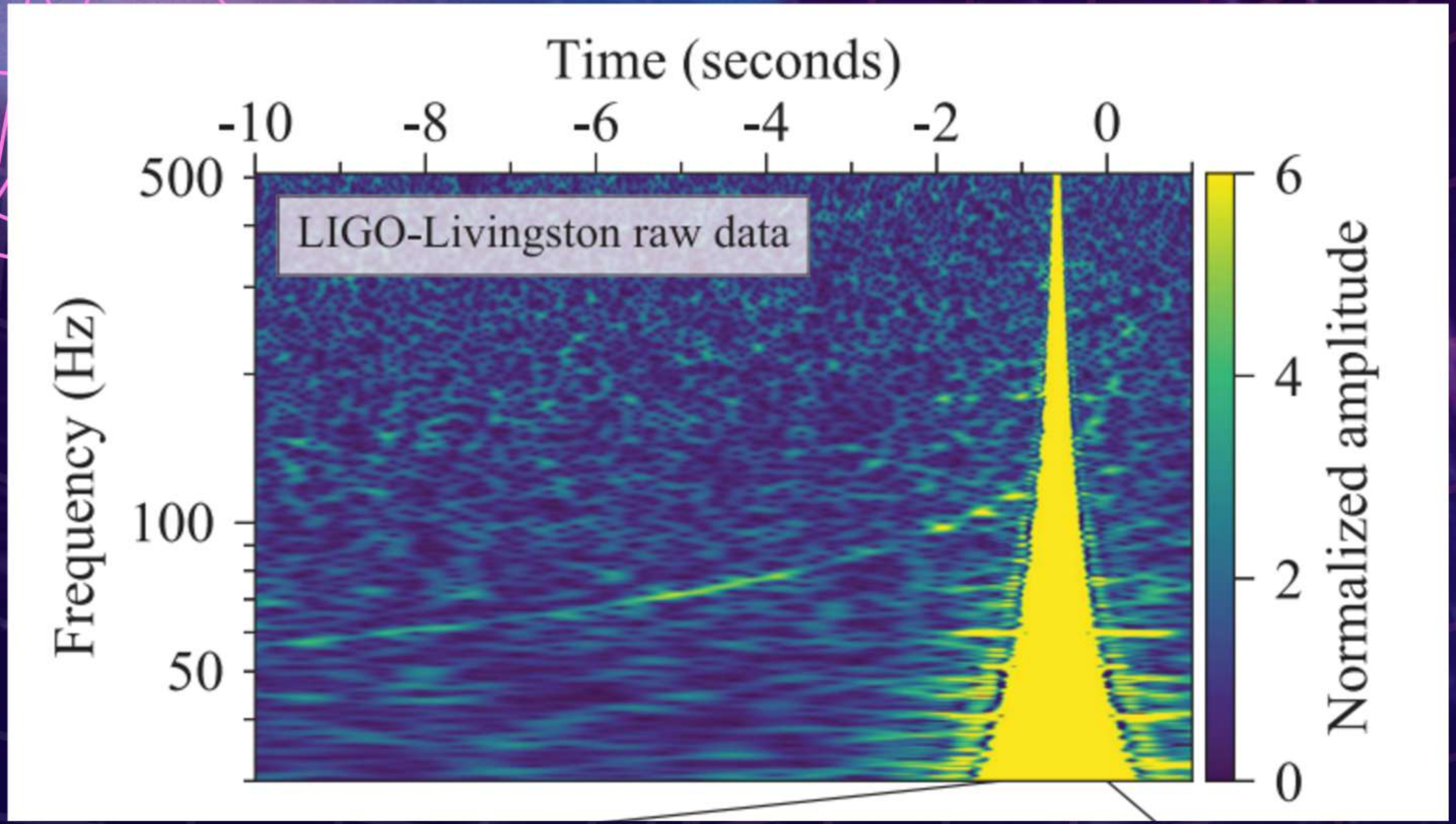
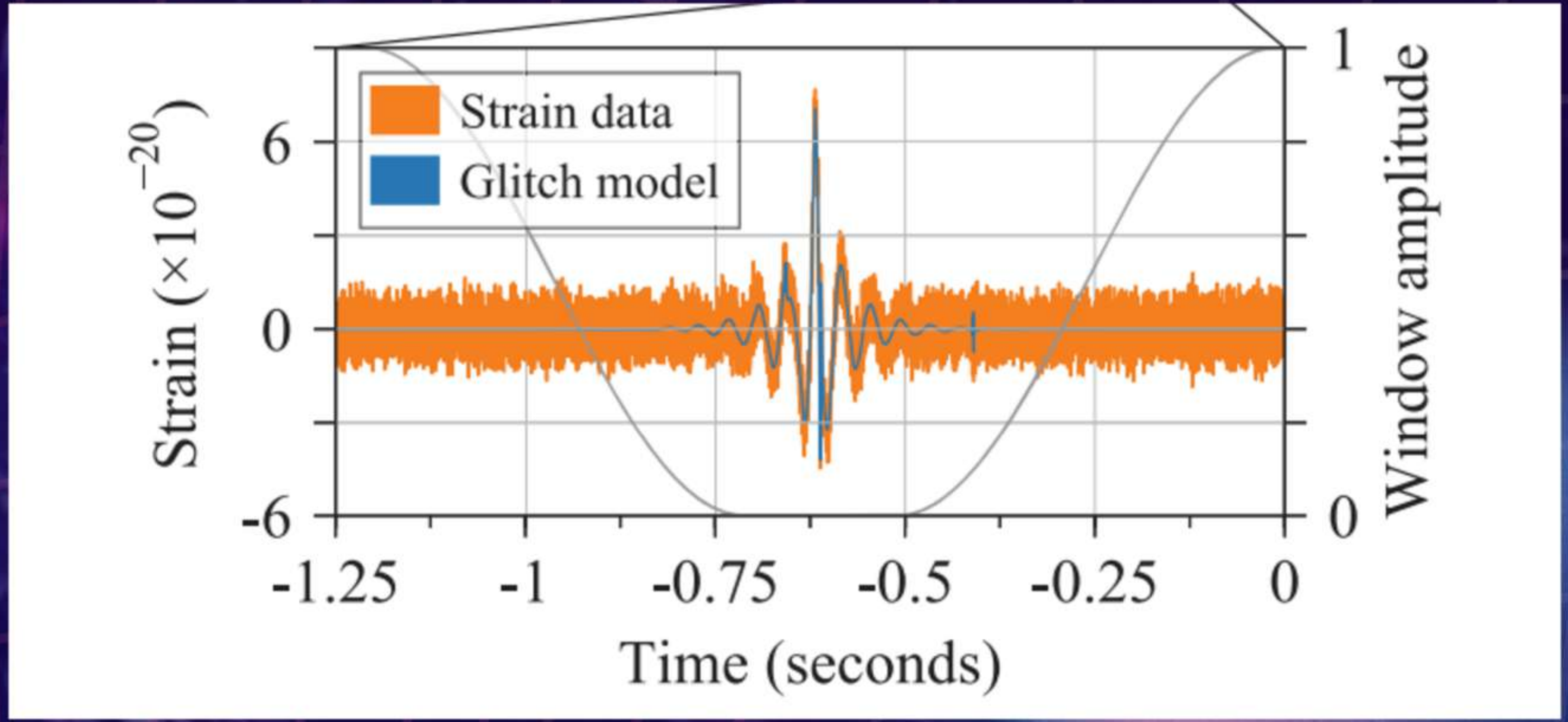


Image from Gravity Spy

Instrumental glitches:

The case of GW170817





- **Common glitch morphology**
- **Happens every few hours**
- **Unknown cause**

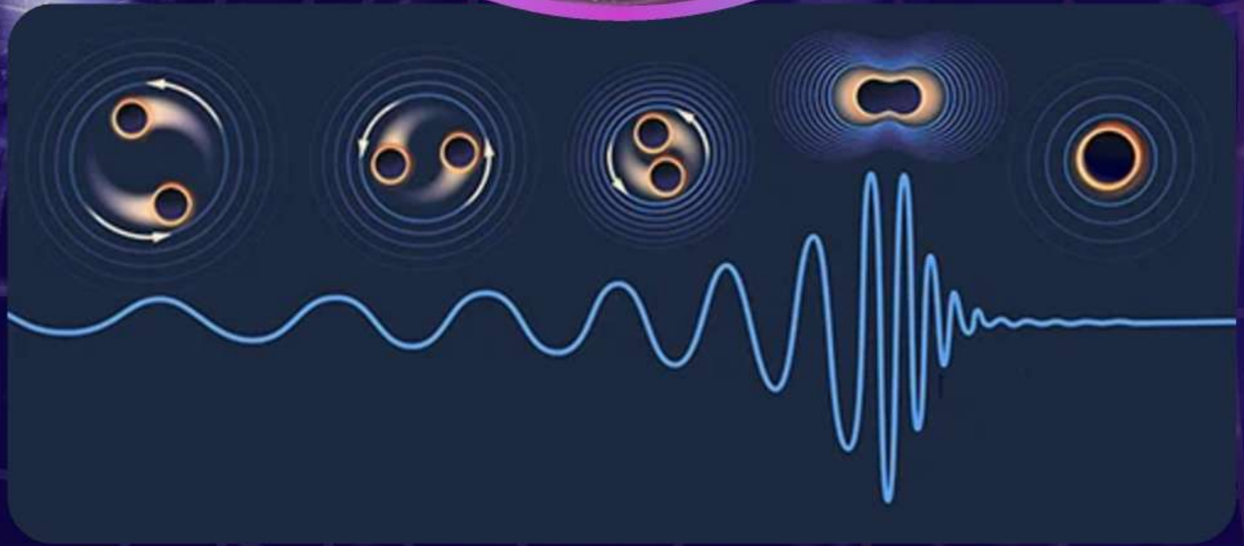
Long signal, short glitch.
WE GOT LUCKY!

Figures from Phys. Rev. Lett. 119, 161101 (2017)

Searching for bursts:

Modelled and Unmodelled

VS

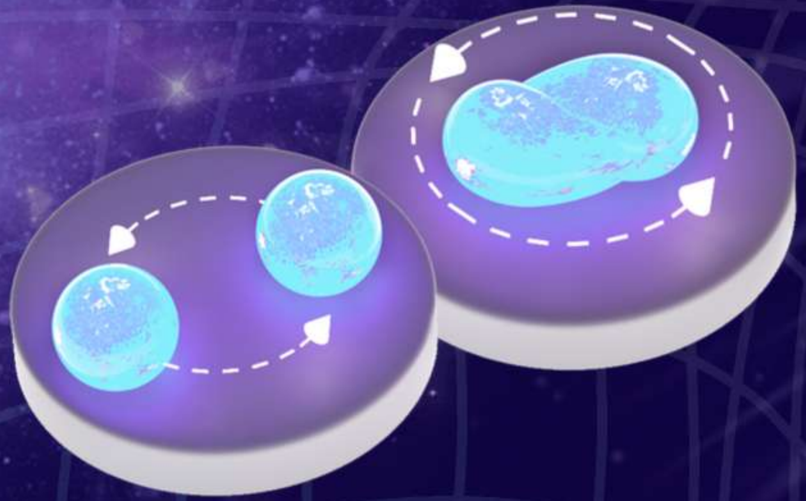


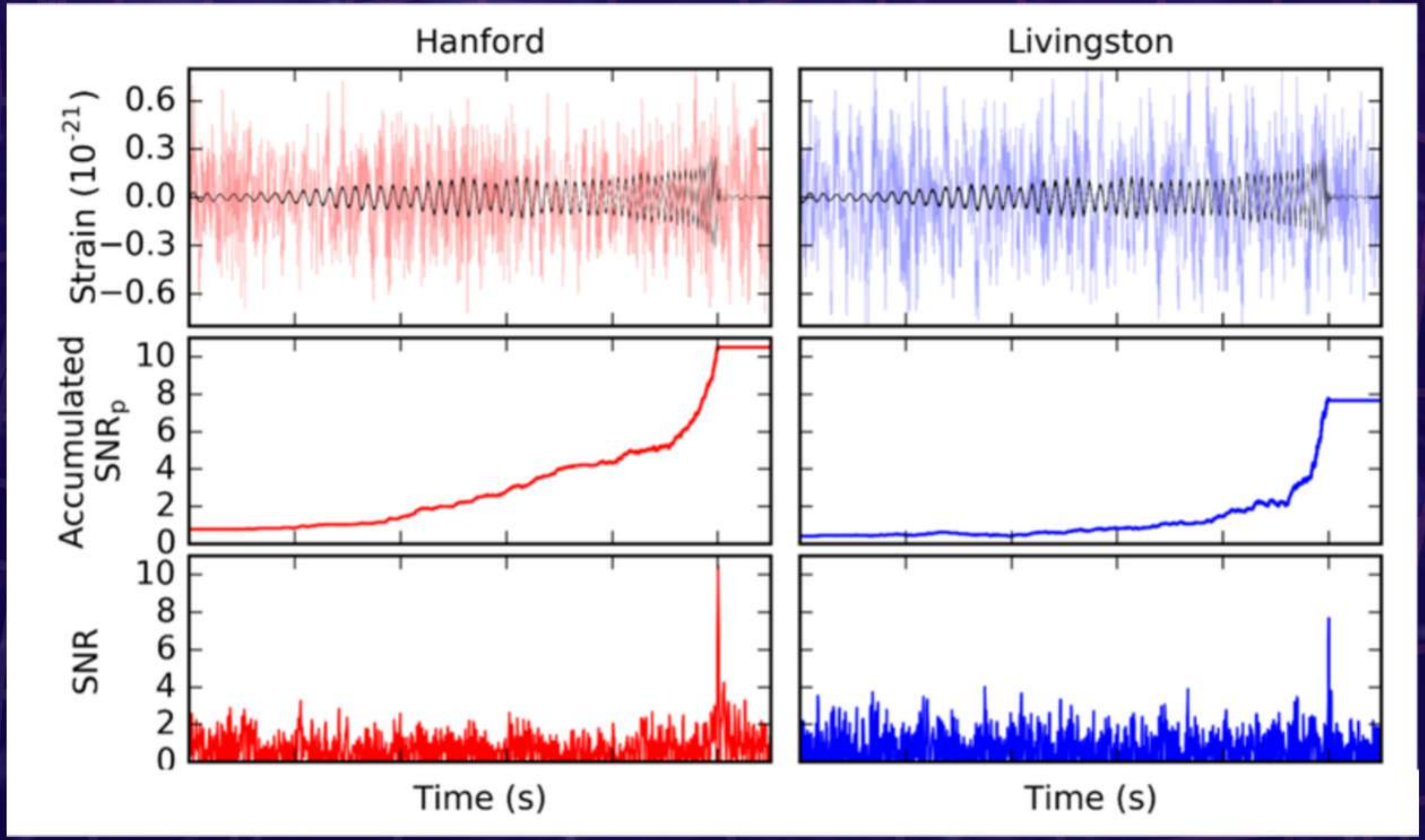
Searching for **Gravitational waves**

(If we know what we're looking for…)

MATCHED-FILTERING

Compare data with large number of waveform models (templates)





Phys. Rev. Lett. 116, 241103, (2016)

Best match template + data

Accumulation of SNR

SNR as we shift the template in time

What if we don't know what we're looking for?

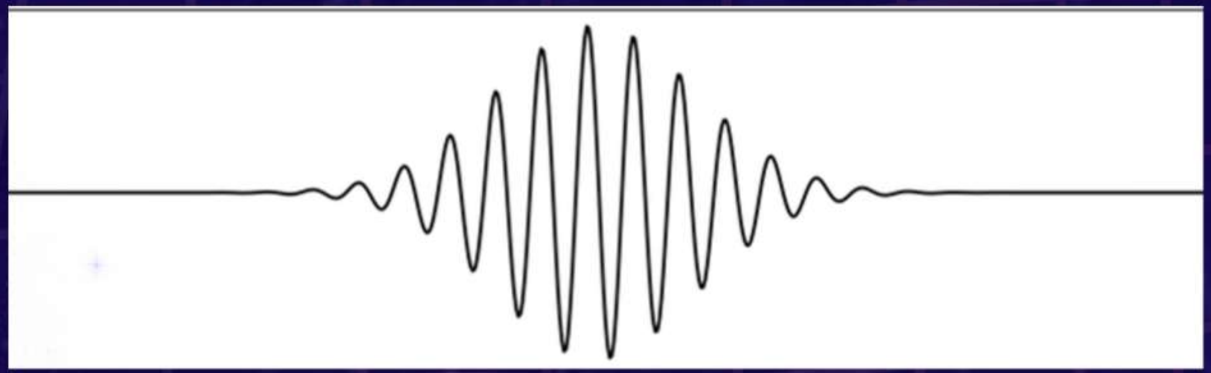
Unmodelled searches

No prior knowledge

Make **minimal assumptions** about the source and signal morphology

Flexible reconstruction of excess power

Tracks time-frequency evolution using **generic frame functions** (e.g. sine-Gaussian wavelets)



Relies strongly on signal coherence

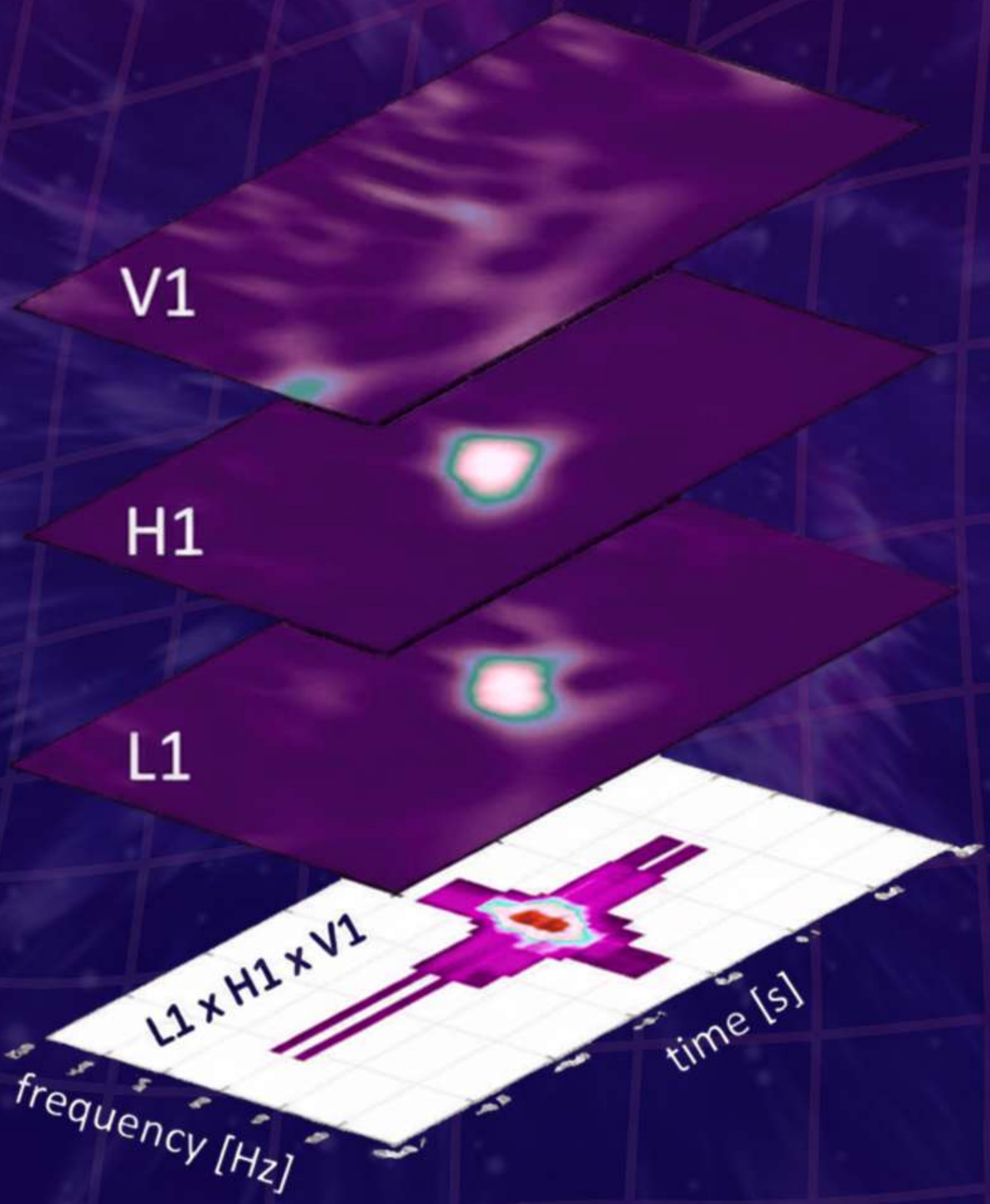
- Increased flexibility = **Increased susceptibility to glitches**
- Cross-correlate data from multiple detectors to distinguish signals from glitches

Examples of

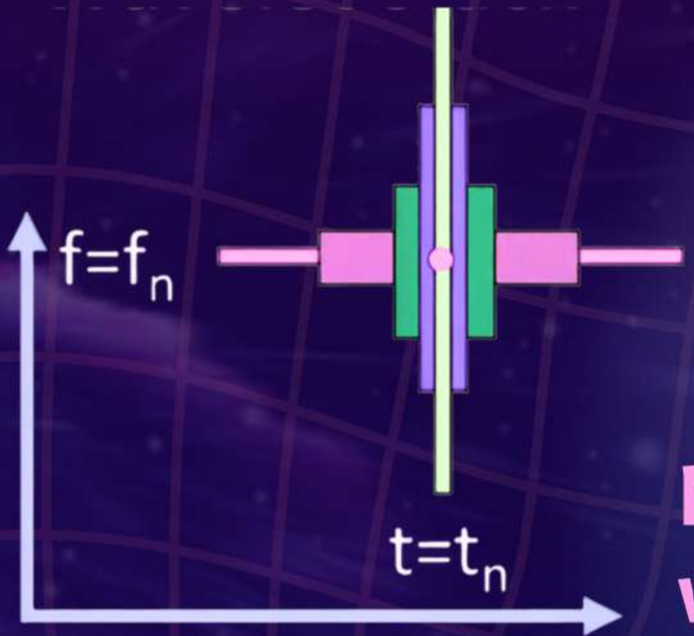
Unmodelled reconstructions

coherent WaveBurst

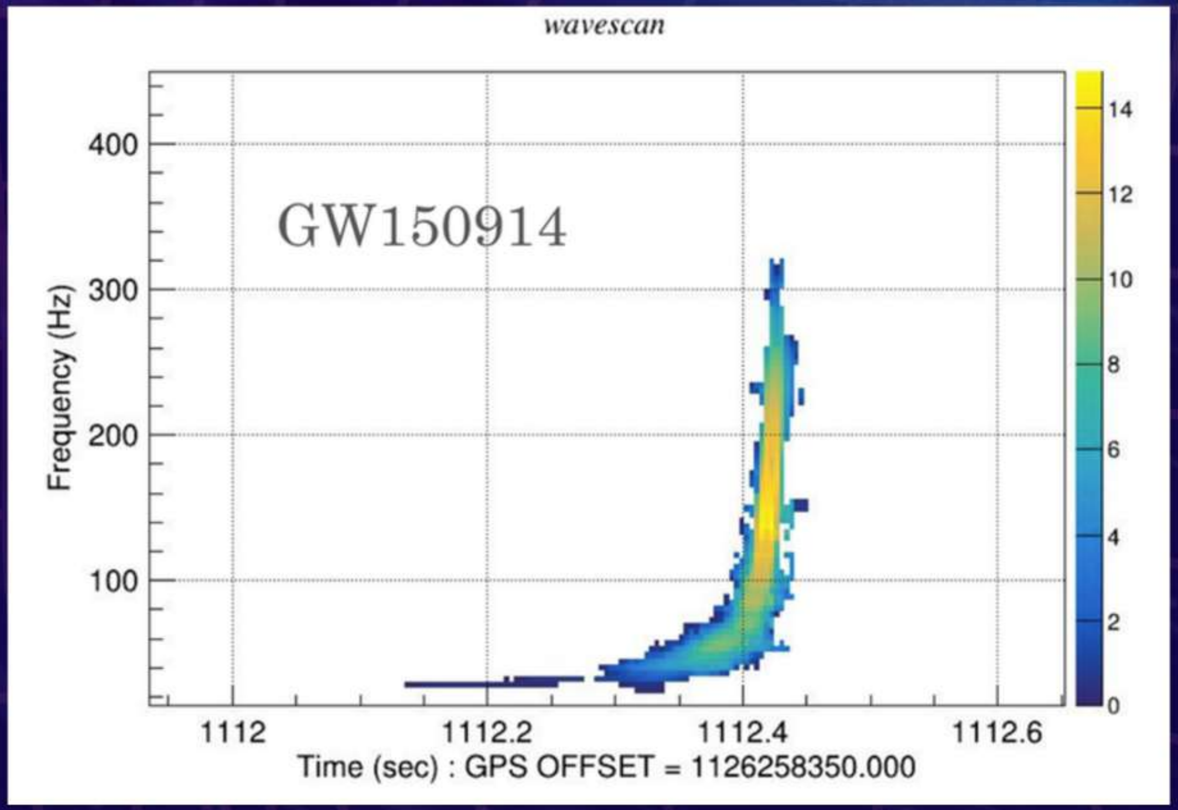
an open source software for gravitational-wave data analysis



Time-frequency map reconstruction



Multi-resolution wavelet stacking

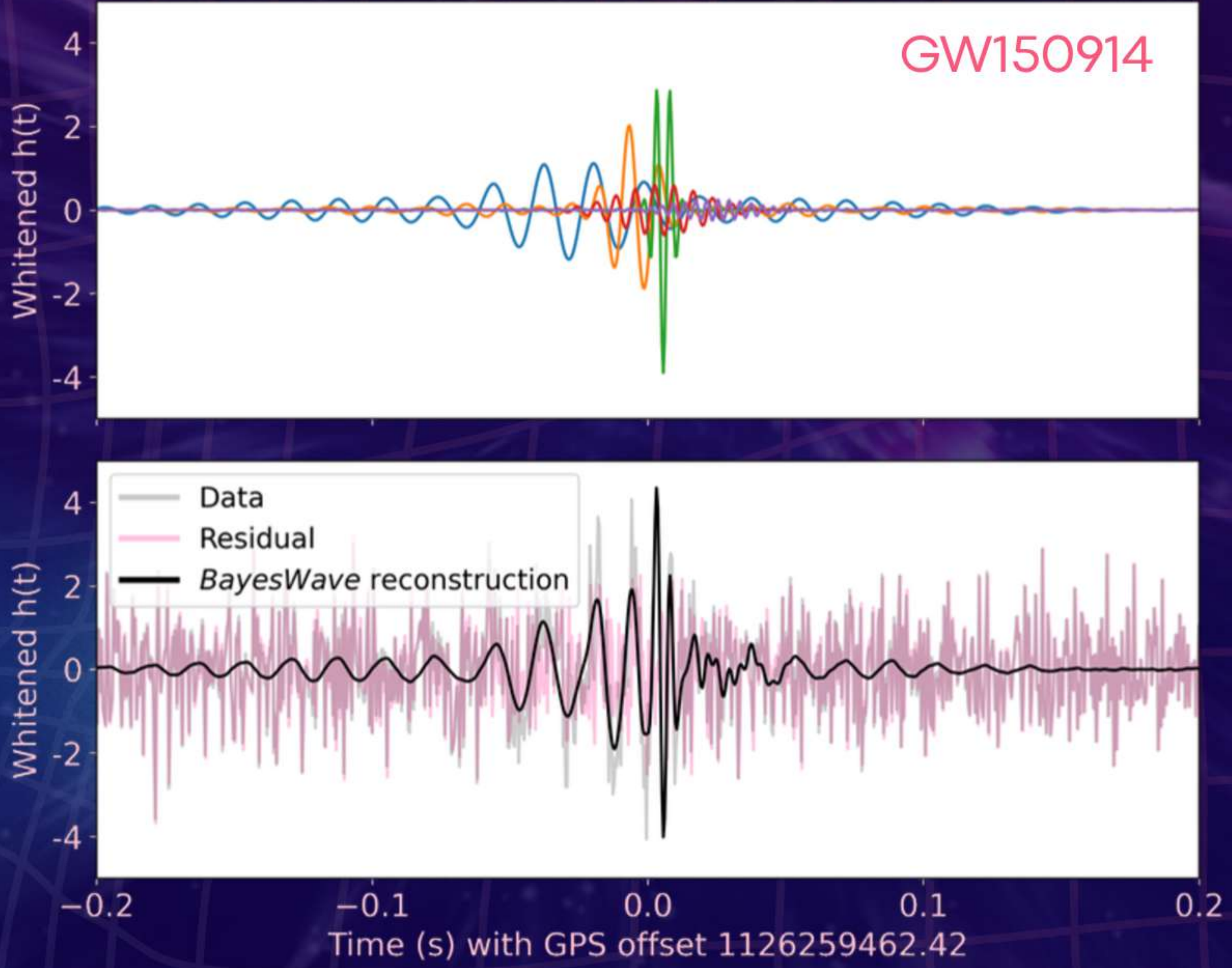


Figures by Sergey Klimenko and Marek Szczepańczyk

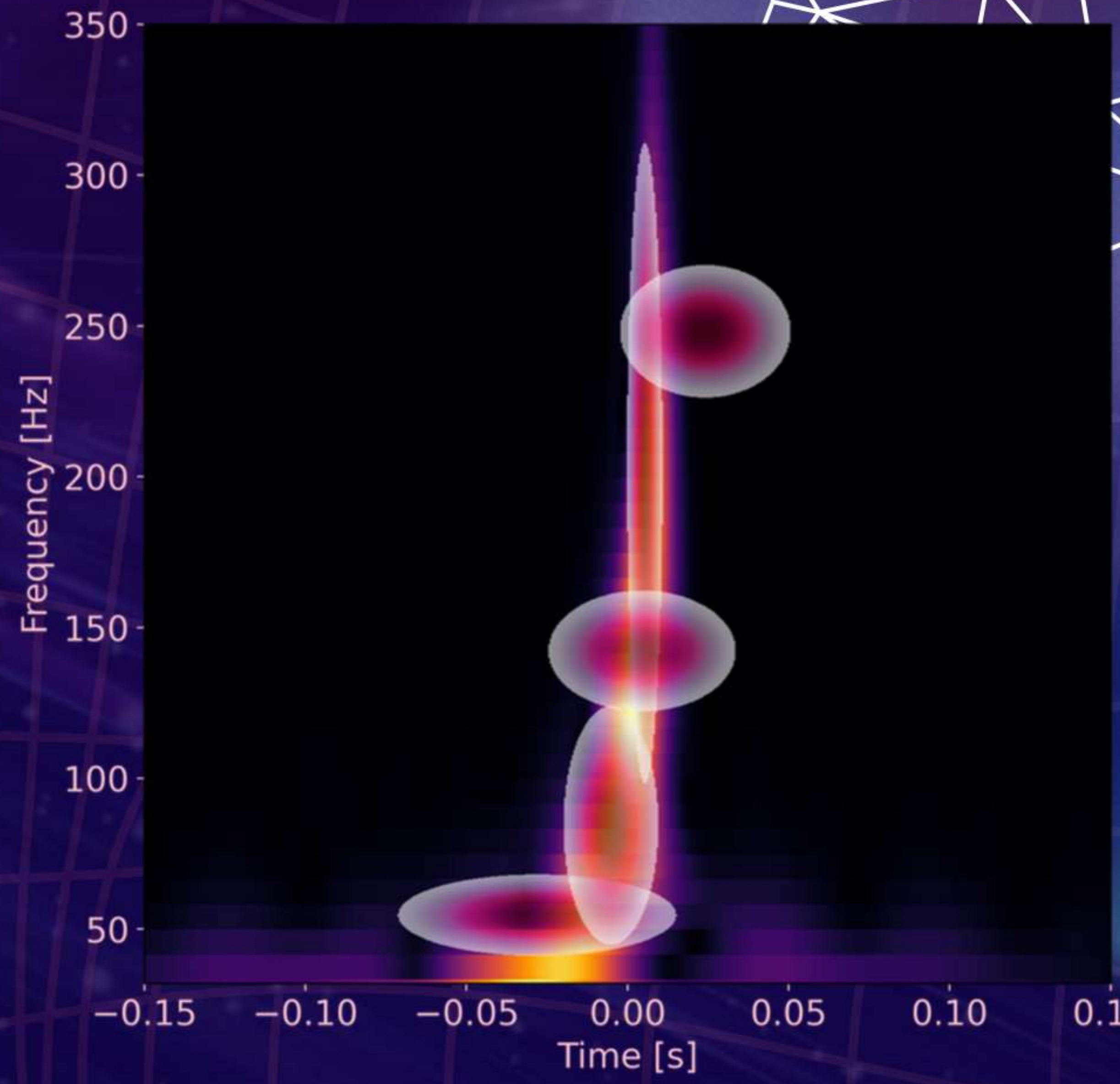
Examples of

Unmodelled reconstructions

BayesWave



Time-domain representation:
sine-Gaussian wavelet reconstruction



Time-frequency representation

A confidence measure:

False Alarm Rate (FAR)

How do we know if a trigger is a SIGNAL or a GLITCH?

How likely is the trigger a noise event?

ASSIGN DETECTION STATISTIC » **IDENTIFY TRIGGER** » **COMPUTE FALSE ALARM RATE**

ASSIGN DETECTION STATISTIC

Numerical score quantifying the significance of an event

IDENTIFY TRIGGER

Identify events with detection statistic exceeding a nominal threshold

COMPUTE FALSE ALARM RATE

FAR [Units: 1/yr]

No. of events per unit time, typically years

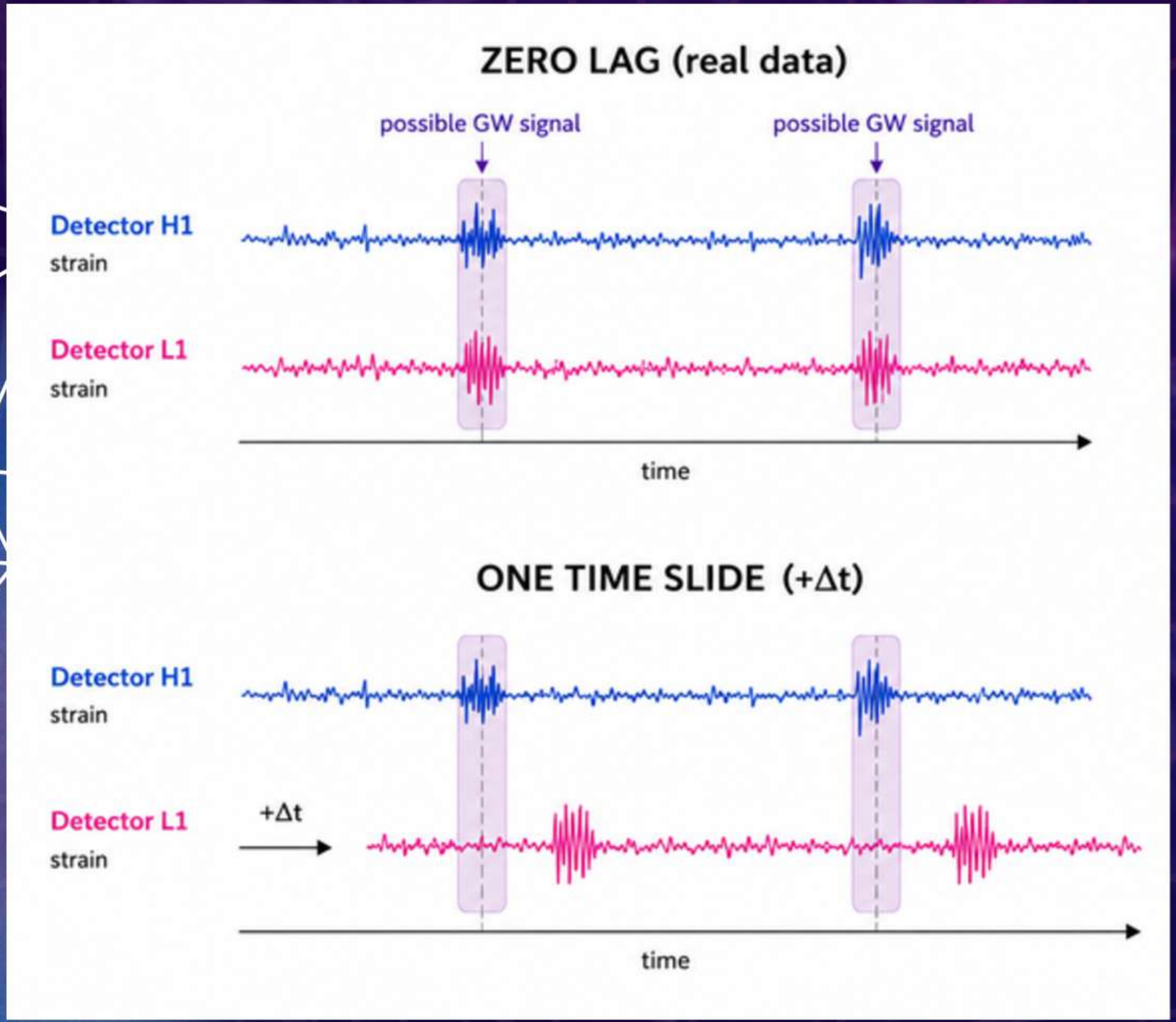
BUT MORE COMMONLY...

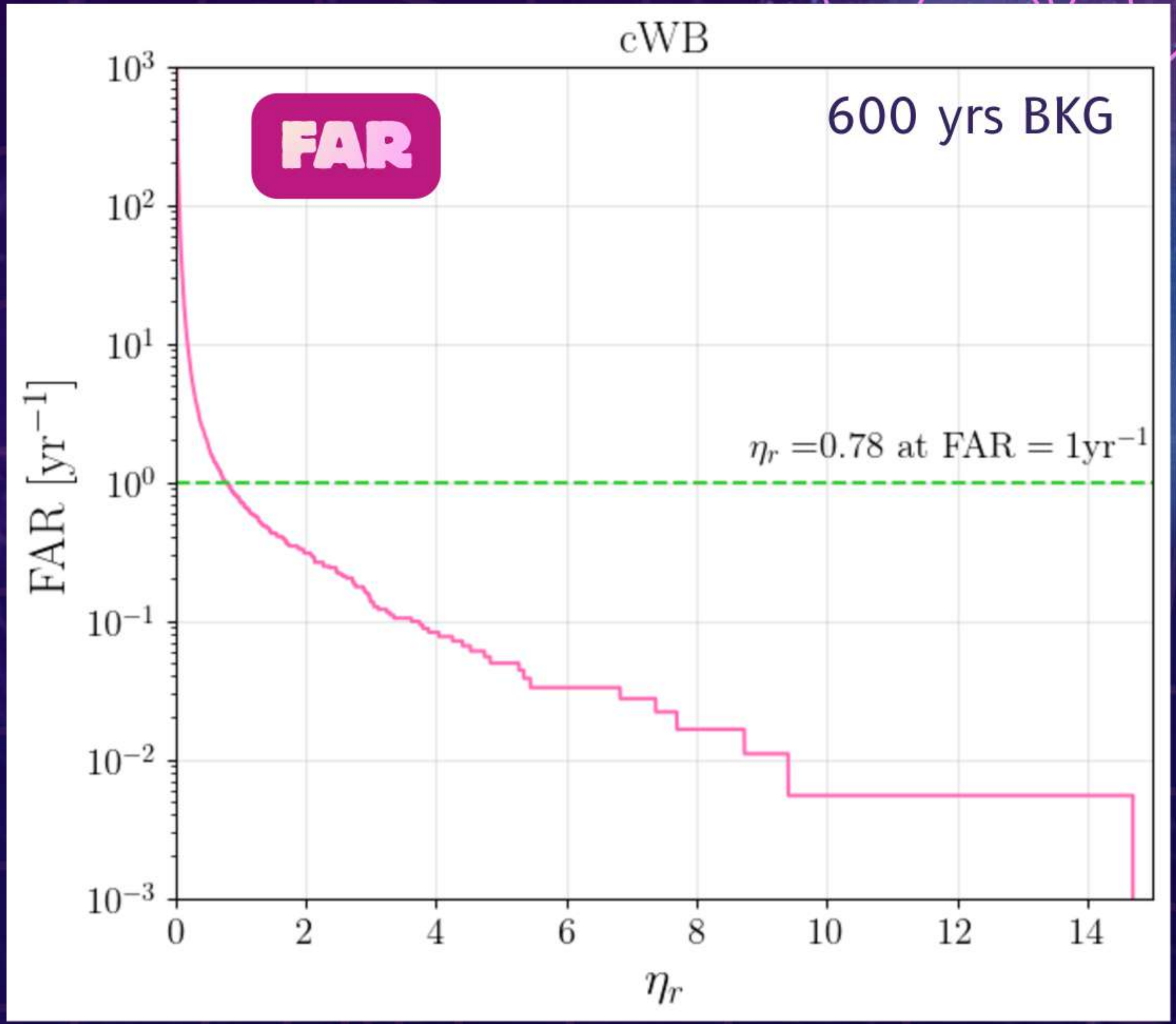
iFAR [Units: yr]

inverse FAR i.e. time per false alarm

Background noise distribution:

Measuring FAR

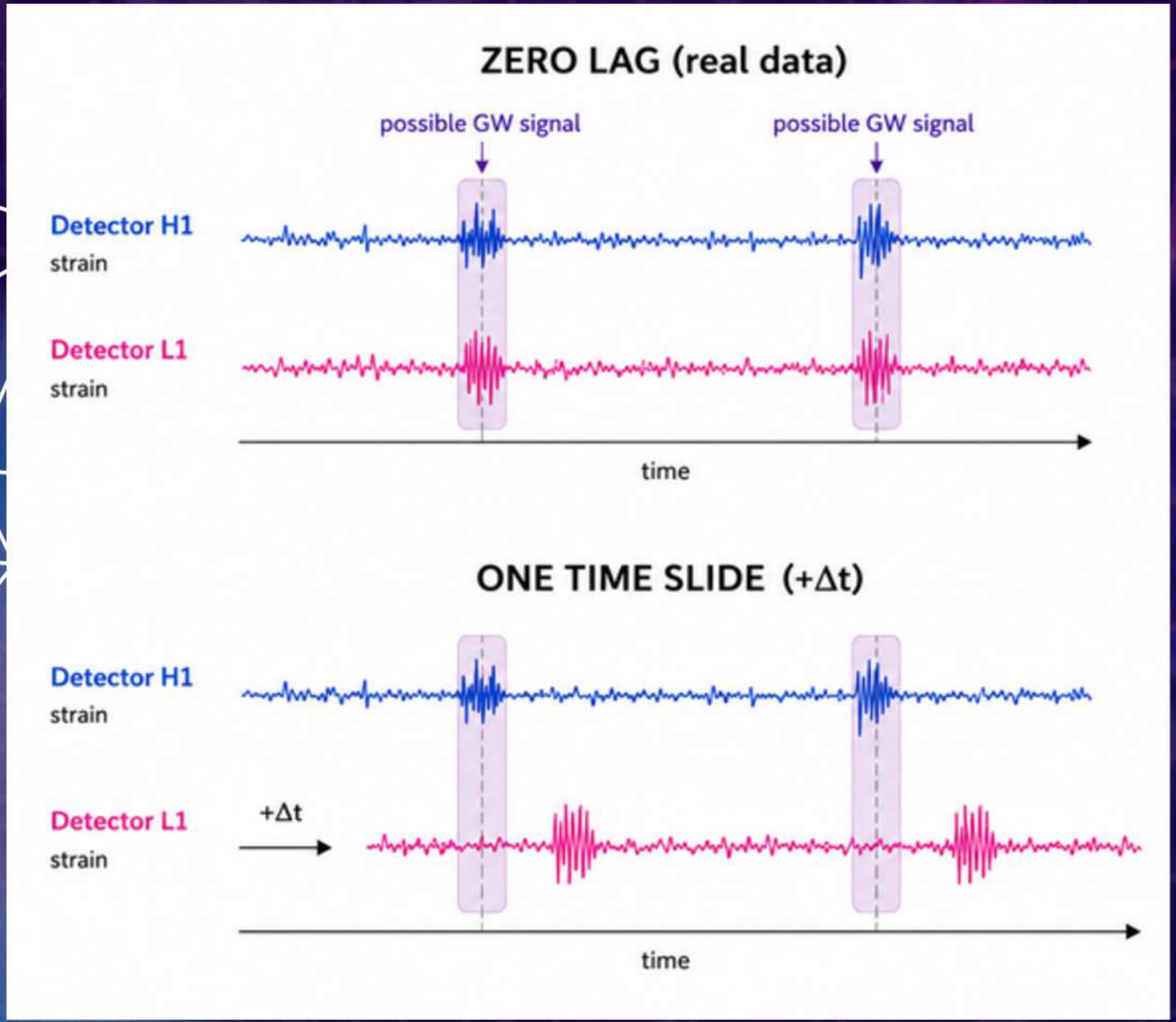


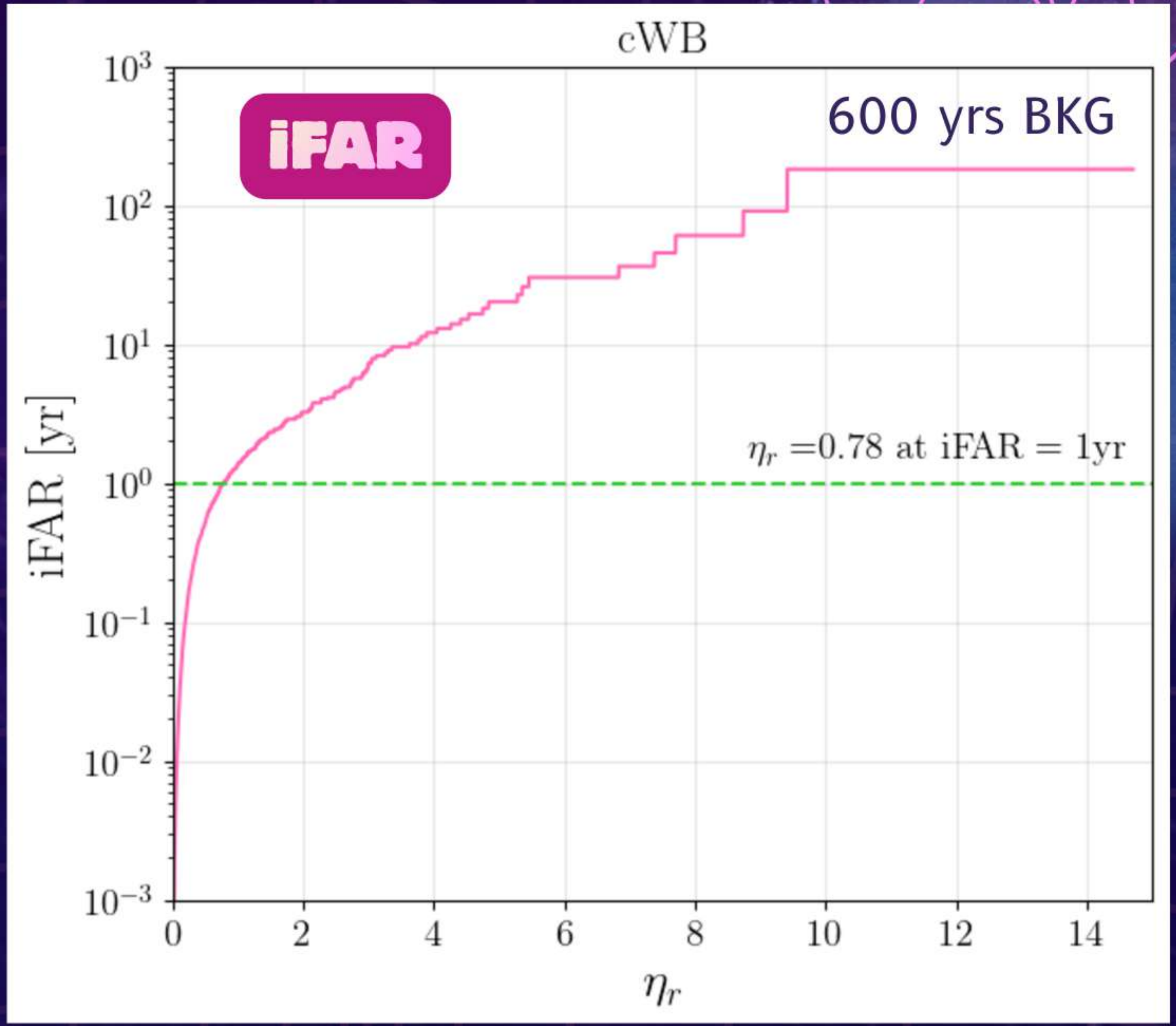


- **Timeslides** break coherence between real signals
- Repeat many times - generating **artificial** extended **background data with NOISE ONLY (~1000 years)**

Empirically measure distribution of detection statistic

Background noise distribution:
Measuring FAR





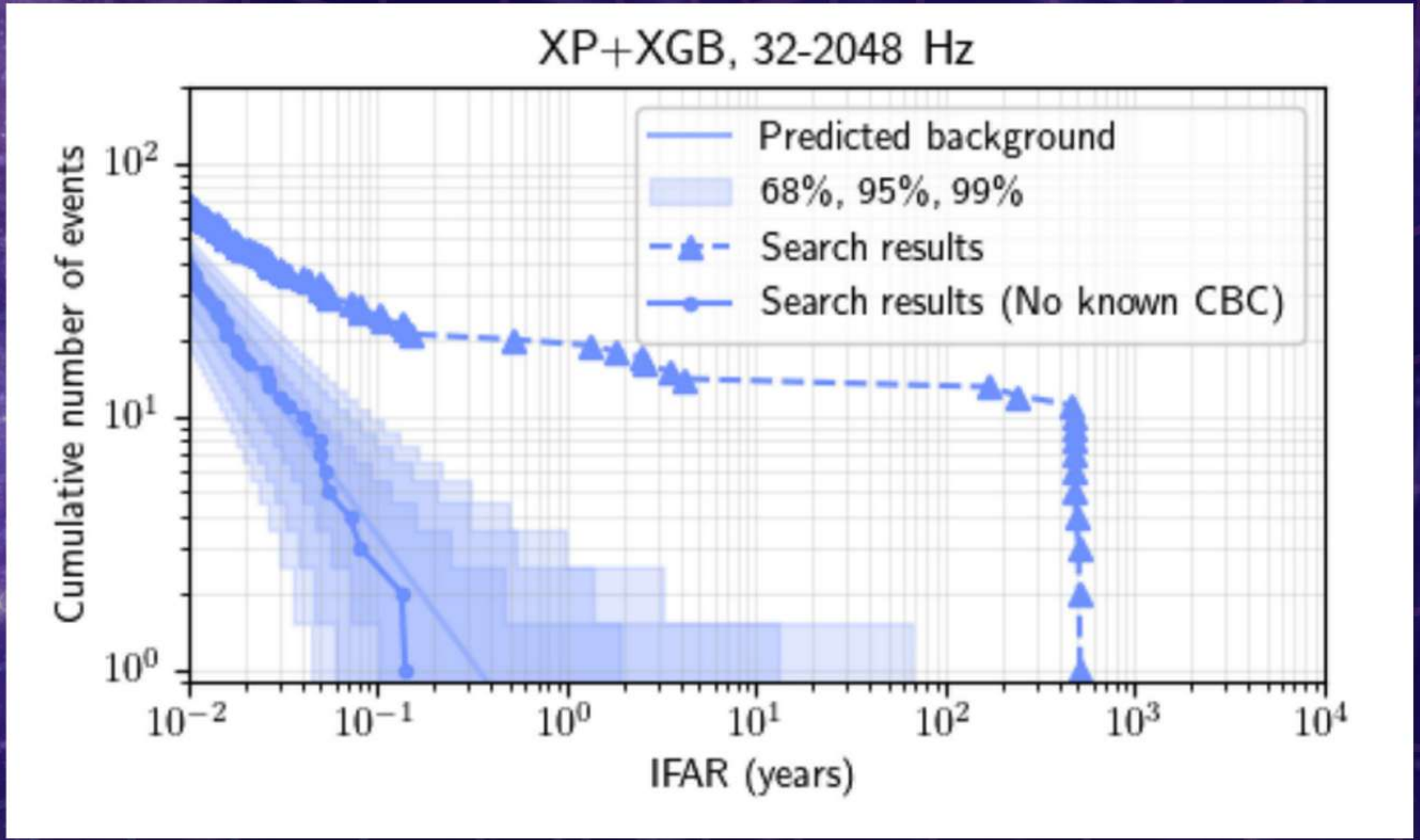
- <u>Timeslides</u> break coherence between real signals
- Repeat many times - generating artificial extended <u>**background data with NOISE ONLY**</u>

Empirically measure distribution of detection statistic

Distribution in real data

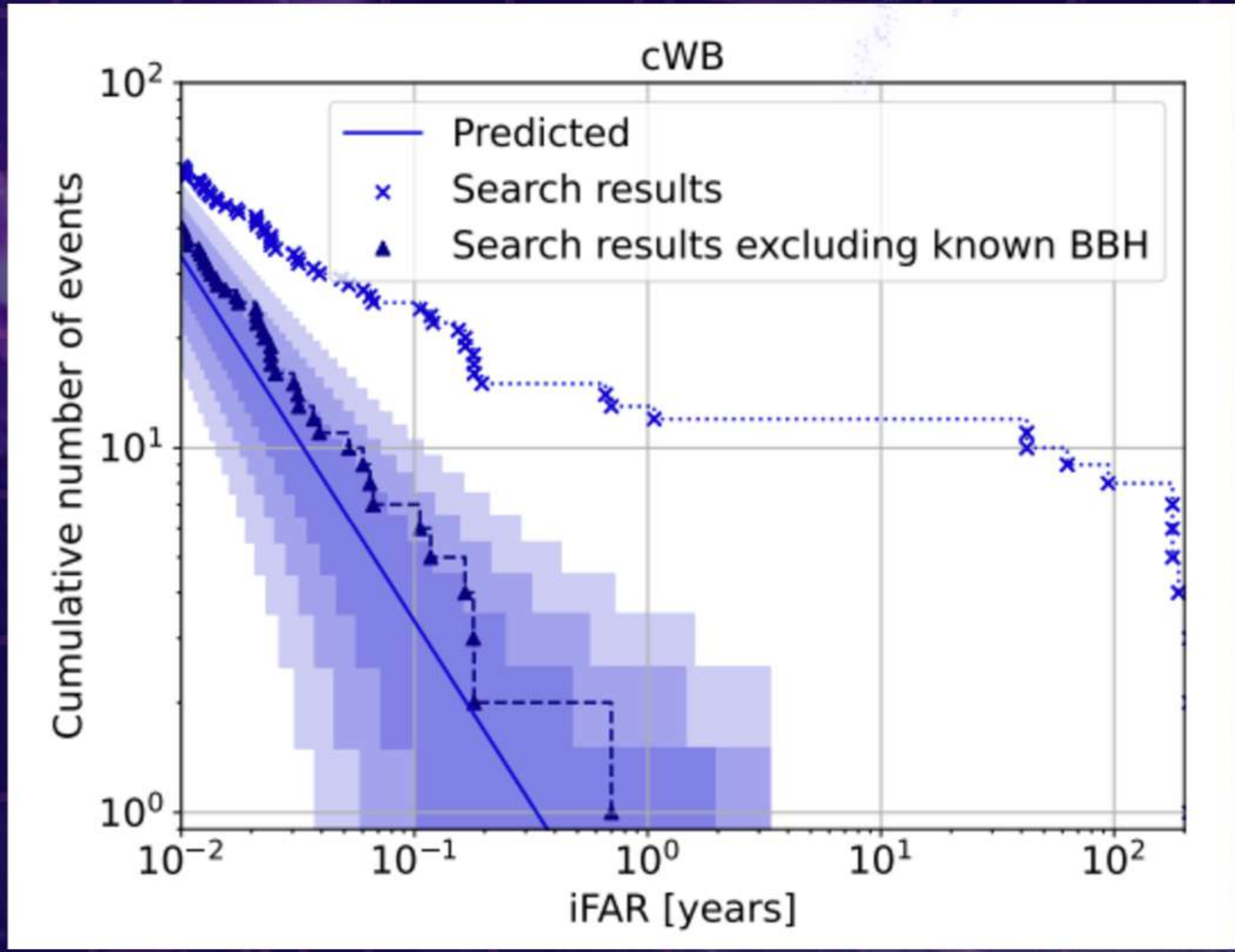
O4a all-sky burst searches

Short-duration
(milliseconds - few seconds)



Phys. Rev. D 112, 102005 (2025)

Long duration
(~1-1000s)



Phys. Rev. D 113, 082004 (2026)

A quick guide to...

GW burst jargon

Types of unmodelled bursts

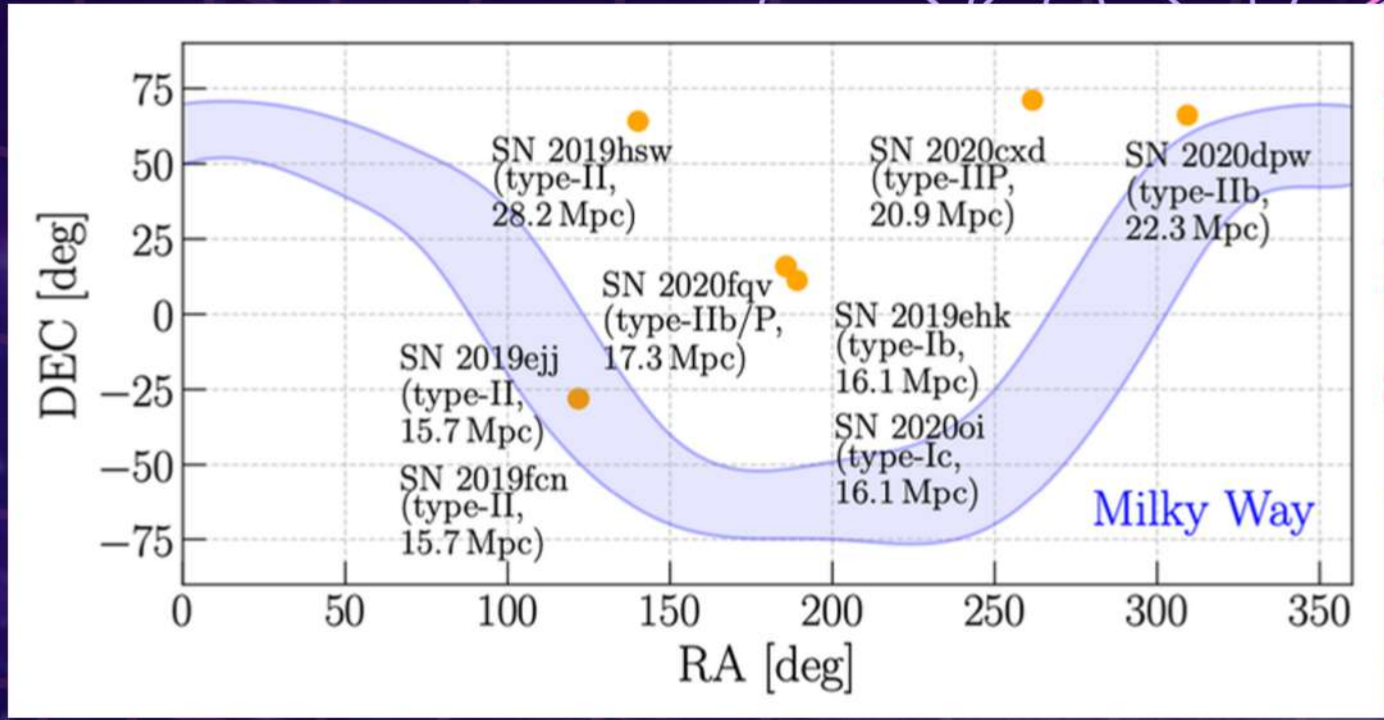
Short vs. Long

	Signal duragion	O3/O4 search pipelines	Anticipated sources (Examples)
SHORT-DURATION BURSTS	Up to a few seconds ($\lesssim 1$ s)	1. **coherent WaveBurst** (cWB) 2. ***BayesWave***	• Core-collapse supernova • Pulsar glitches • Fast Radio bursts • Gamma-ray bursts* • Cosmic string cusps
LONG-DURATION BURSTS ($\gtrsim 1$ s)		1. **coherent WaveBurst** (cWB) 2. **PySTAMPAS**	• Non-axisymmetric deformations in magnetars • Eccentric binaries • Black-hole accretion disk instability

Types of searches

All-sky vs. targeted

	ALL-SKY	TARGETED
Goal	Detect unknown or unexpected GW bursts	Search for GW emission associated with a known astrophysical event
Search region	Entire sky, all distances	Source location and distance
Search duration	Entire observing run	“On-source” window (expected signal time interval)
Source type	N/A	Core-collapse supernovae, FRBs, GRBs, pulsar glitches...



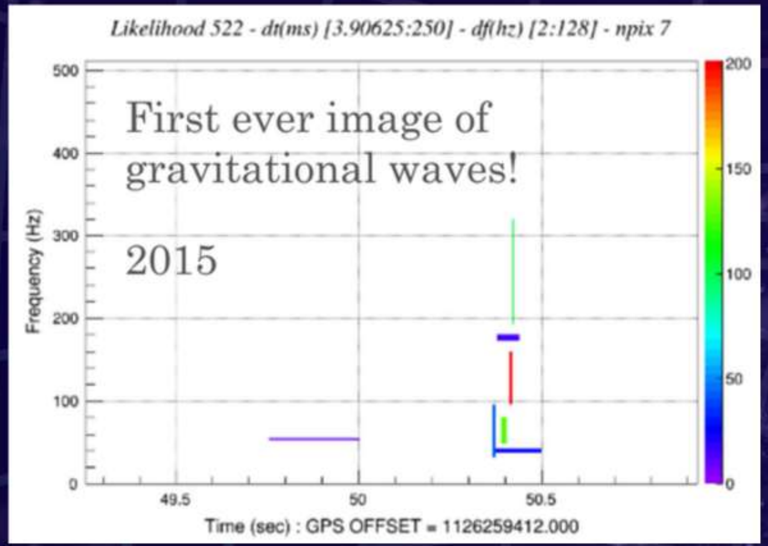
Supernova	And+17 s11	Kur+16 s15	Kur+22 s50	Mez+20 C15
SN 2019ehk	-	6.57	-	0.52
SN 2019ejj	0.78	7.94	1.67	1.73
SN 2019fcn	0.58	7.40	0.80	0.84
SN 2019hsw	0.70	5.60	1.82	2.24
SN 2020oi	0.63	6.53	-	1.15
SN 2020cxd	**0.88**	**8.90**	**2.13**	**2.74**
SN 2020dpw	0.79	8.66	1.70	2.46
SN 2020fqv	0.73	6.86	1.56	2.38

Phys. Rev. D 110, 042007 (2024)

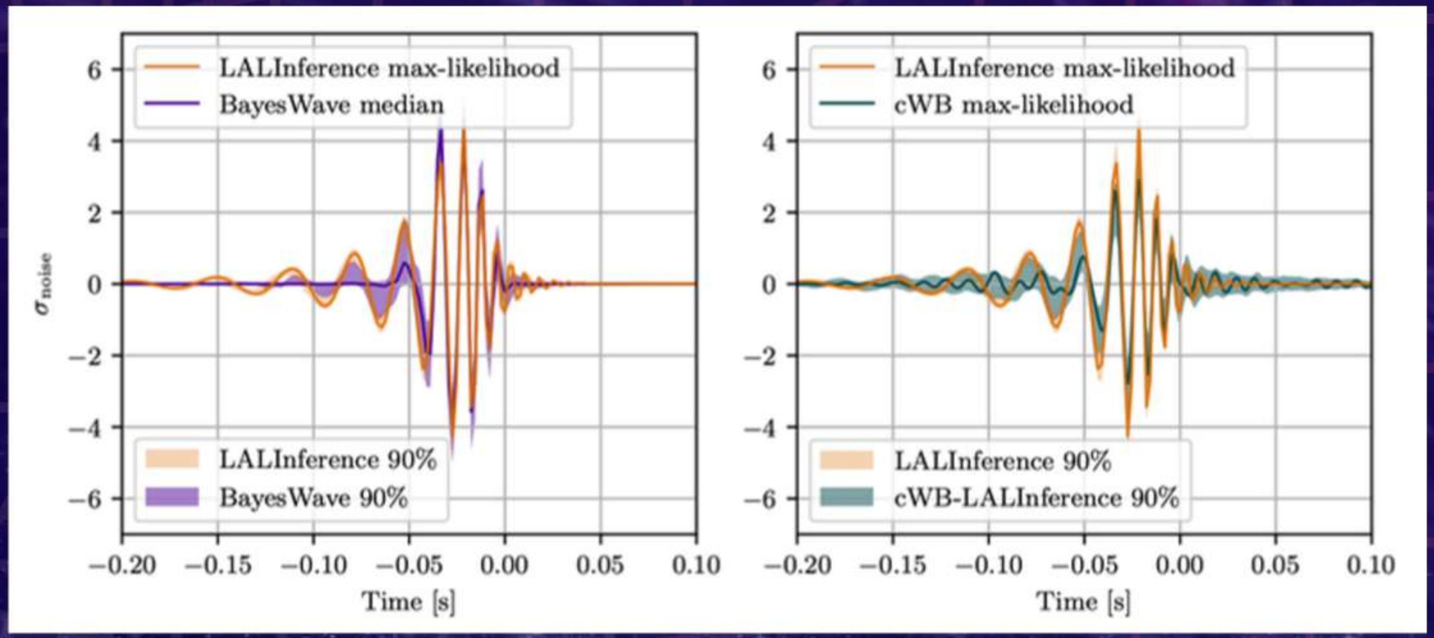
Role of Burst pipelines

Low-latency searches

cWB is used in low-latency (real-time) searches, complementing template-based searches.

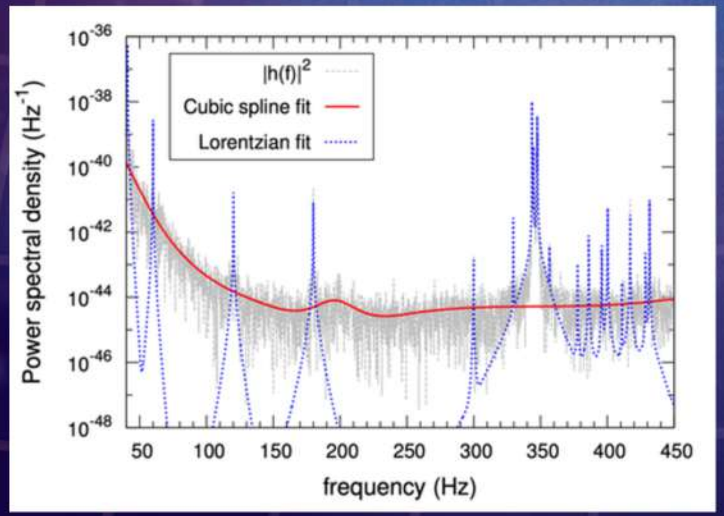


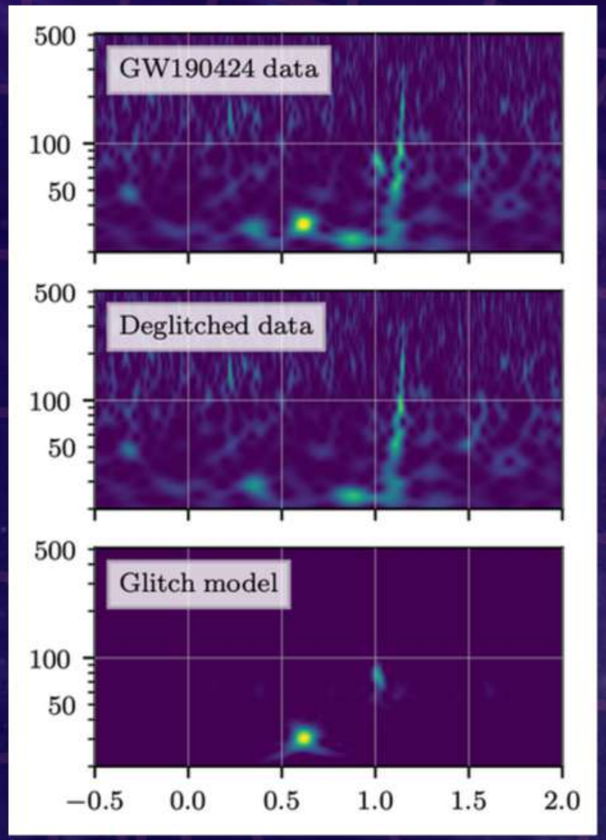
Waveform consistency tests



Spectral noise estimation

BayesWave models Gaussian noise for template-based parameter estimation





Glitch subtraction

BayesWave can simultaneously model a signal and glitch, making it an effective glitch subtraction tool.

Figures left to right from:
[1] Phys. Rev. D 93, 122004 (2016)
[2] GWTC-2: Phys. Rev. X 11, 021053 (2021)
[3] Class. Quantum Grav. 39 245013 (2022)
[4] *BayesLine*: Phys. Rev. D 91, 084034 (2015)

Talk Summary

BURSTS VS. GLITCHES

GW bursts: transient, non-Gaussian and coherent excess power

Instrumental glitches: Mimic or mask GW burst signals

GW BURST ANALYSIS

Burst searches: Flexible reconstruction with minimal assumptions

False alarm rates: Measures the confidence of a detection

No unmodeled burst detections to date...but burst searches still play key roles in LVK

LVK burst publications

Questions?

Come chat with me!

- Gravitational-wave data analysis
- Core-collapse supernovae
- Radio pulsar glitch detection
- Bayesian inference + Hidden Markov models
- University of Melbourne open-source codebases
- Science communication and teaching

Contact:
christine.yishuen@unimelb.edu.au

SCAN ME

CURIOUS ABOUT MY RESEARCH?